DocuQuery Corporation: Annual Technical & Financial Report 2024

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1. Executive & Financial Summary

DocuQuery Corporation delivered strong operational and financial performance during fiscal year 2024. Total revenue increased from $100M in 2023 to $115M in 2024, representing an annual revenue growth rate of 15.0%. Through disciplined infrastructure scaling and local model execution, Operating Expenses decreased by 8.0%, falling from $50M to $46M.

Financial Metric	FY 2023	FY 2024	Growth / Margin
Total Revenue	$100M	$115M	+15.0%
Operating Expenses	$50M	$46M	-8.0%
Operating Income	$20M	$25M	25.0% Margin
EBITDA	$24M	$30M	30.0% Margin

2. Applied AI Architecture & Retrieval Mechanisms

DocuQuery Engine implements a state-of-the-art hybrid Graph-RAG architecture. For dense semantic representation, the platform runs BAAI/bge-small-en-v1.5 locally via FastEmbed on CPU, yielding 384-dimensional dense vectors. Graph traversal is driven by KùzuDB, an embedded Python property graph database supporting 1-to-2 hop multi-entity traversal. Dense vector similarity and graph connectivity are combined using Reciprocal Rank Fusion (RRF) with constant k=60. Candidates are subsequently reranked using FlashRank running ms-marco-TinyBERT-L-2-v2 with sub-10ms latency.

3. Deterministic AST Numerical Verification

Unlike conventional LLM chatbots that hallucinate tabular figures, DocuQuery Engine introduces a deterministic Python AST verification layer. Extracted arithmetic claims (such as growth rates, ratios, and operating margins) are mapped directly to raw Markdown tables ingested from source PDFs. A restricted Python Abstract Syntax Tree (AST) parser evaluates mathematical formulas and attaches a Verified Calculation badge.

4. Operational Subscription Growth

User subscriptions rose from 10,000 to 12,000 active enterprise accounts, reflecting an expansion growth rate of 20.0%. The platform enforces strict PDF upload limits of 20MB per file and provides cascading hard deletion across PostgreSQL, pgvector chunks, and KùzuDB graph tables.

Product Tier	2023 Accounts	2024 Accounts	Growth Rate
Enterprise Workspaces	10,000	12,000	+20.0%
Research Nodes	2,500	3,500	+40.0%